Ordinary Least Squares

Lecture 2 (Stata code)

Table of contents

Reading . . . 2

Data **3**
LCF (2023) . . . 3

Estimation: An intuitive approach **3**
Guess 1 . . . 4
Guess 2 . . . 6
Structure . . . 6
Fit . . . 7
Aggregation . . . 8
Minization . . . 8

Ordinary Least Squares **8**
Problem . . . 8
First order conditions . . . 9
Simple case . . . 9
Sample Analogue . . . 11
Decomposition . . . 11
Fitted values . . . 11
Interpretation . . . 12

Example: LCF (2023) **12**
Estimation . . . 12
Decomposition . . . 12
Fitted values . . . 13
Residual . . . 14
Binary regressor: . . . 14
Summary stats . . . 14

OLS - dummy variable . . . 15
Sample analogue . . . 16
Multivariate models . . . 16
OLS - multivariate . . . 16
Interpretation . . . 17
Partialling out . . . 17
Demonstration . . . 18

Goodness-of-Fit **19**
Measure of fit . . . 20
Chasing $\mathbf{R}^2$ . . . 20
Adjusted $\mathbf{R}^2$ . . . 21
Example: LCF (2023) . . . 21
Adding "irrelevant" variables . . . 21

Does OLS work? **22**
Accuracy . . . 23
Random variable . . . 23
Example: LCF (2023) - Version 2 . . . 24
Example: LCF (2023) - Version 3 . . . 24
Example: LCF (2023) - Version 4 . . . 24

References **25**
Bibliography . . . 25

Reading

This lecture covers material from Wooldridge (2025):

- Chapter 2.2 - "Deriving the Ordinary Least Squares Estimates"
- Chapter 2.3 - "Properties of OLS on Any Sample of Data"
- Chapter 2.4 - "Units of Measurement and Functional Form" (covered in Lecture 1)
- Chapter 3.2 - "Mechanics and Interpretation of Ordinary Least Squares"
- Chapter 6.3 - "More on Goodness-of-Fit and Selection of Regressors"

The lecture does not cover Chapter 2.6 "Regression through the Origin and Regression on a Constant". I would suggest reading the section, but we will not have time to cover it.

Data

For each observation we observe:

$$[y_i, x_{i1}, x_{i2}, ..., x_{ik}] \quad i = 1, ..., n$$

Stacked together, we have a matrix of data:

$$\begin{bmatrix} y_1 & x_{11} & x_{12} & ... & x_{1k} \\ y_2 & x_{21} & x_{22} & ... & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{n1} & x_{n2} & ... & x_{nk} \end{bmatrix}$$

Just like in Stata/R

LCF (2023)

Suppose you wanted to estimate the the marginal propensity to consume in the UK using data from the Living Costs and Food Survey:

- a *household level* dataset with information on consumption, income, and other characteristics.

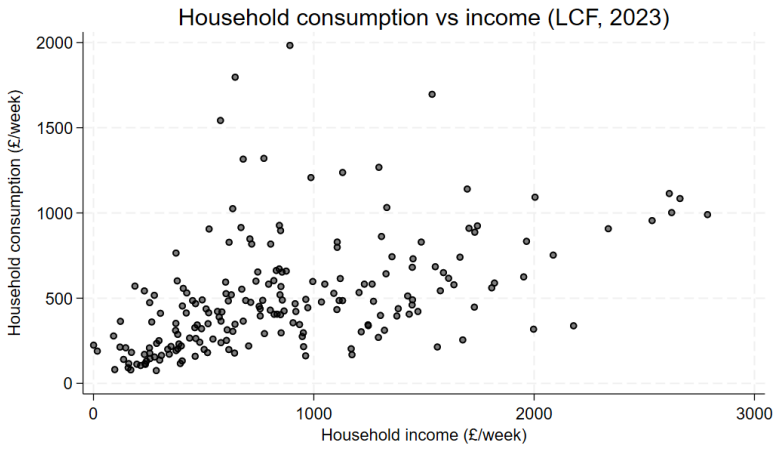
$$[cons_i, inc_i, children_i, size_i, tv_license_i, month_i] \quad i = 1, ..., n$$

Estimation: An intuitive approach

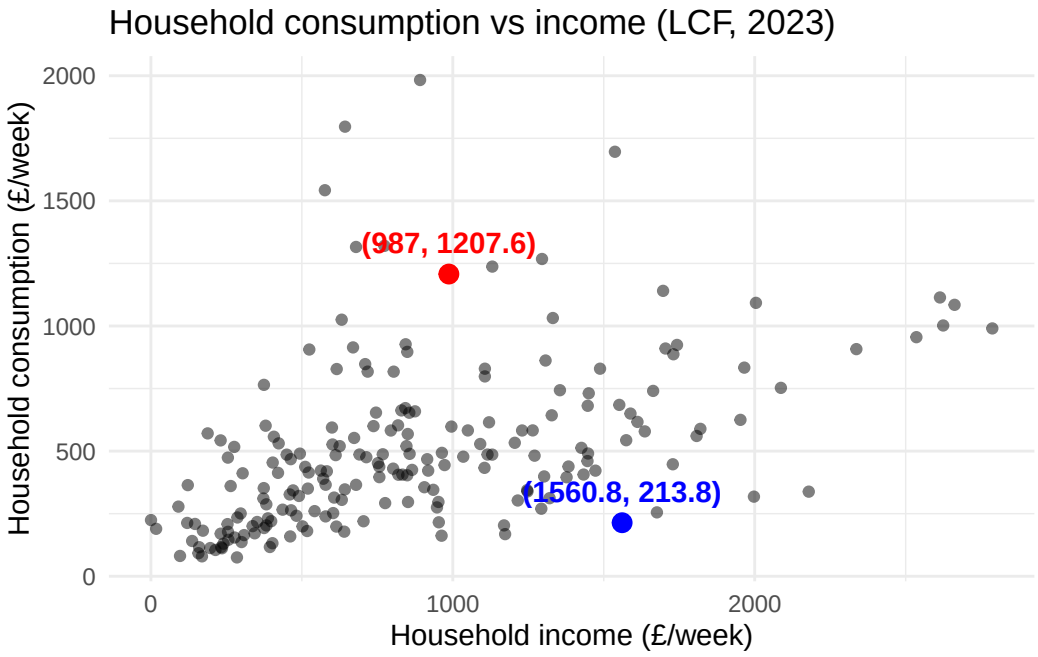
Does the relationship between consumption and income look linear?

```
* Open data
use "$data_dir\lcf_2023.dta", clear

* Plot
scatter hhcon hhinc, mc(black) mfc(%50) ytitle("Household consumption (£/week)") xtitle("Hous
```

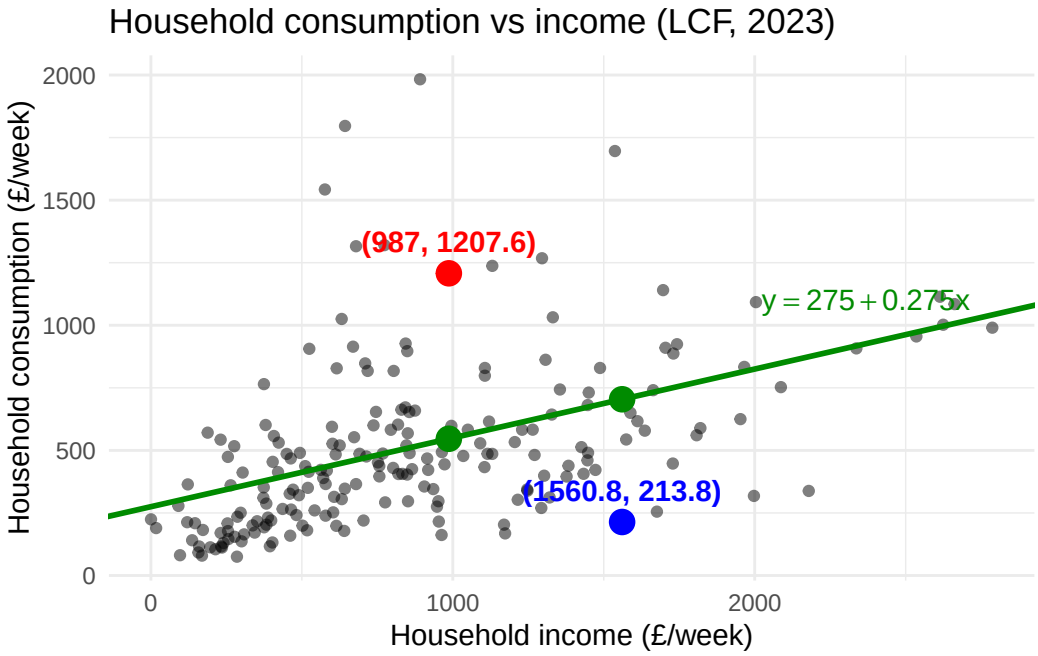


Consider the following two observations:

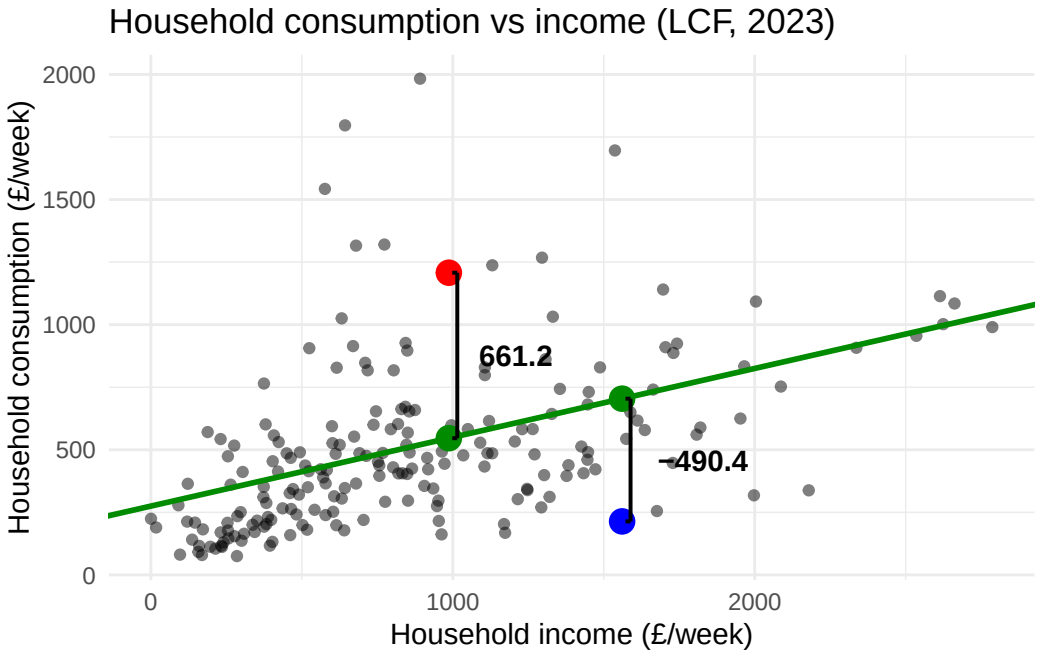


Guess 1

Suppose we plot a line through the points:

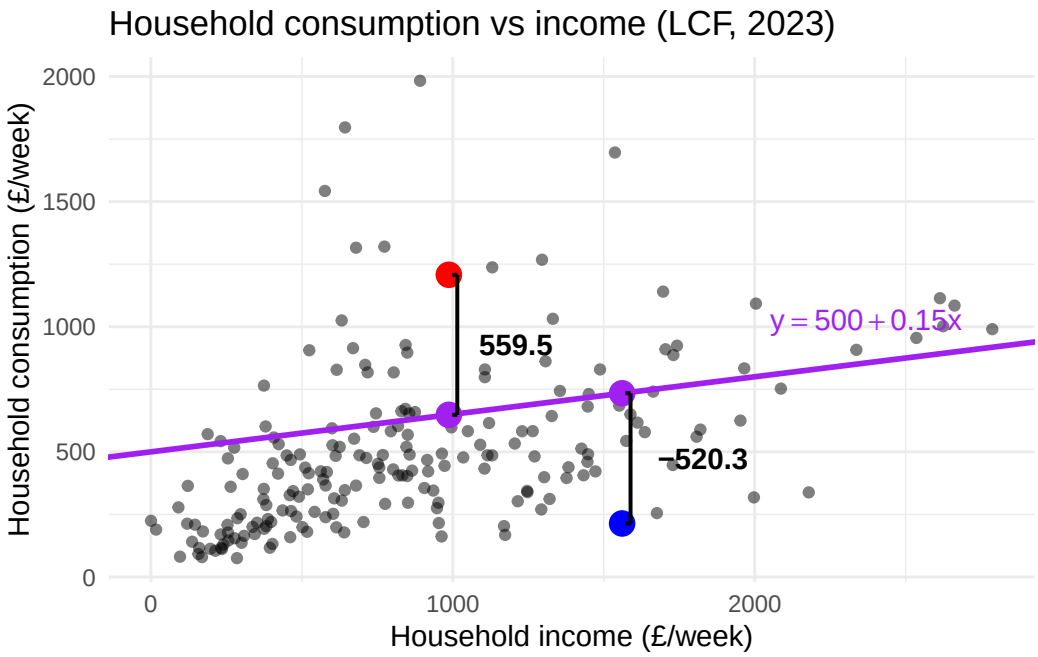


The vertical distance between points in the data and the line tells us how the difference between the "prediction" of the line (for a given inc_i) and the actual data.



Guess 2

What if we chose a different plot?



Which guess was better? Why?

Structure

Let's formalize the above process:

- **Goal:** we want to pick the line that best fits the data.
- **Problem:** what do we mean by "fit"?
  - For a given value of inc_i in the data, the "fitted" value of $cons_i$ is

$$\widehat{cons}_i = b_0 + b_1 inc_i$$

where $[b_0, b_1]$ are the chosen parameters.

- For example,
  - $\widehat{cons}_i = 275 + 0.275 inc_i$
  - $\widehat{cons}_i = 500 + 0.15 inc_i$

Fit

We need a measure of distance:

Candidate 1: difference

$$cons_i - \widehat{cons}_i$$

- **PROBLEM**: could be negative (i.e. not a measure of distance)

Candidate 2: absolute value

$$|cons_i - \widehat{cons}_i|$$

- **PRO**: intuitive
- **CON**: difficult to work with (i.e. not globally differentiable)

Candidate 3: squared deviation

$$(cons_i - \widehat{cons}_i)^2$$

- always positive
- amplifies bigger deviations, which is more *efficient* (more on that soon!)

Aggregation

What we care about is the difference between the prediction and data across all observations:

$$\sum_{i=1}^{n}(cons_i - \widehat{cons}_i)^2$$

And since $\widehat{cons}_i = b_0 + b_1 inc_i$

$$\sum_{i=1}^{n}(cons_i - b_0 - b_1 inc_i)^2$$

Minization

The value of these squared deviations (from the prediction) depend on $[b_0, b_1]$. It seems intuitive then to pick the values that minimize this value:

$$\min_{b_0, b_1} \quad \sum_{i=1}^{n}(cons_i - b_0 - b_1 inc_i)^2$$

Ordinary Least Squares

Problem

Minimization problem:

$$\hat{\beta} = \arg\min_{b} \quad \sum_{i=1}^{n}(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik})^2$$

where $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k]$ and $b = [b_0, b_1, \ldots, b_k]$.

- hence the name "Least Squares"

First order conditions

Differentiate w.r.t. b_j's and then set to 0. For $\hat{\beta}_0$ (the constant):

$$\begin{aligned} 0 &= 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \quad 0 &= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \quad \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}_1 - \cdots - \hat{\beta}_k \bar{x}_k \end{aligned}$$

> **Note**
>
> The inclusion of a constant in the model ensures that the residual is always mean zero:
>
> $$0 = \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = \sum_{i=1}^{n}\hat{u}_i$$

For $\hat{\beta}_j \quad j = 1, \ldots, k$

$$\begin{aligned} 0 &= 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})x_{ij} \\ \Rightarrow \quad 0 &= \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})x_{ij} \end{aligned}$$

There are k of these first-order conditions (plus the 1 for the constant)

$\Rightarrow k + 1$ equations and $k + 1$ unknowns.

Simple case

In the case of a single regressor, the solution is more easily solved.

- For $\hat{\beta}_0$ (the constant):

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- For $\hat{\beta}_1$:

$$
\begin{aligned}
& 0 = 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i \\
\Rightarrow \quad & 0 = \sum_{i=1}^{n}(y_i - (\bar{y} - \hat{\beta}_1\bar{x}) - \hat{\beta}_1 x_i)x_i \qquad \text{substitute in} \quad \hat{\beta}_0 \\
\Rightarrow \quad & 0 = \sum_{i=1}^{n}((y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x}))x_i
\end{aligned}
$$

The next step exploits the fact that:

$$\sum_{i=1}^{n}\hat{u}_i = 0 \Rightarrow \bar{x}\sum_{i=1}^{n}\hat{u}_i = 0 \Rightarrow \sum_{i=1}^{n}\hat{u}_i\bar{x} = 0$$

Therefore we can minus 0 from the right-hand side without changing anything.

$$
\begin{aligned}
& 0 = \sum_{i=1}^{n}(\underbrace{(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})}_{\hat{u}_i})x_i - \sum_{i=1}^{n}\hat{u}_i\bar{x} \\
\Rightarrow \quad & 0 = \sum_{i=1}^{n}((y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x}))(x_i - \bar{x}) \\
\Rightarrow \quad & \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)}
\end{aligned}
$$

Linear algebra

The above expression - along with its multivariate counterpart - has a very nice linear algebra counterpart. The vector of OLS estimators (including the intercept) is given by,

$$\hat{\beta} = (X'X)^{-1}X'Y$$

Where X is a $n \times (k+1)$ matrix of regressors (including vector of 1's for the intercept) and Y a $n \times 1$ vector. It's a very simple expression and demonstrates the value of knowing a little linear algebra. This how you define the OLS estimator in graduate-level modules like EC5207 *Econometric Methods and Applications*.

Sample Analogue

Recall from Lecture 1, under the MLR assumptions (notably, $E[x_i u_i] = 0$)

$$\begin{aligned}\beta_1 =& \frac{Cov(y_i, x_i)}{Var(x_i)} \\ \beta_0 =& E[y_i] - \beta_1 E[x_i]\end{aligned}$$

The OLS estimators for a simple model are the **sample analogue**

$$\begin{aligned}\hat{\beta}_1 =& \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)} \\ \hat{\beta}_0 =& \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

OLS replaces expectations with sample averages.

Decomposition

Given the OLS estimates, we can decompose the outcome into two parts: fitted values ($\hat{y}_i$) and residual ($\hat{u}_i$)

$$y_i = \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\hat{y}_i} + \hat{u}_i$$

Fitted values

You can show,

- $\sum_{i=1}^n \hat{u}_i = 0$: residuals are mean zero
- $\sum_{i=1}^n x_i \hat{u}_i = 0$: residuals and regressors are uncorrelated
- $\sum_{i=1}^n \hat{y}_i \hat{u}_i = 0$: residuals and fitted values are uncorrelated
- $\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i$: fitted values have the same mean as outcome
- $\sum_{i=1}^n y_i \hat{u}_i = \sum_{i=1}^n (\hat{y}_i + \hat{u}_i)\hat{u}_i = \sum_{i=1}^n \hat{u}_i^2$

Some of these are exercises in Tutorial 1

Interpretation

The $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ also gives us the interpration of the estimator:

$$\hat{\beta}_1 = \frac{d\hat{y}_i}{dx_i} \quad \text{OR} \quad \Delta\hat{y}_i = \hat{\beta}_1 \Delta x_i$$

and

$$\hat{\beta}_0 = \hat{y}_i \quad \text{for } x_i = 0$$

Example: LCF (2023)

Estimation

```
* Estimate bivariate linear regression
regress hhcon hhinc
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     57.08
       Model |  4842359.39         1  4842359.39   Prob > F        =    0.0000
    Residual |  16798378.5       198  84840.2954   R-squared       =    0.2238
-------------+----------------------------------   Adj R-squared   =    0.2198
       Total |  21640737.9       199  108747.427   Root MSE        =    291.27

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .2705895   .0358165     7.55   0.000     .1999587    .3412203
       _cons |   268.0611   37.42847     7.16   0.000     194.2515    341.8707
------------------------------------------------------------------------------
```

Decomposition

```
* Generate fitted values and residuals
predict fitted
predict residual, resid
```

```
* Compute averages
sum hhcon fitted residual
```

```
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
       hhcon |        200    504.1663    329.7687      75.16    1983.02
      fitted |        200    504.1663    155.9919    268.1639    1022.213
    residual |        200   -1.28e-07    290.5408   -519.6953    1473.709
```

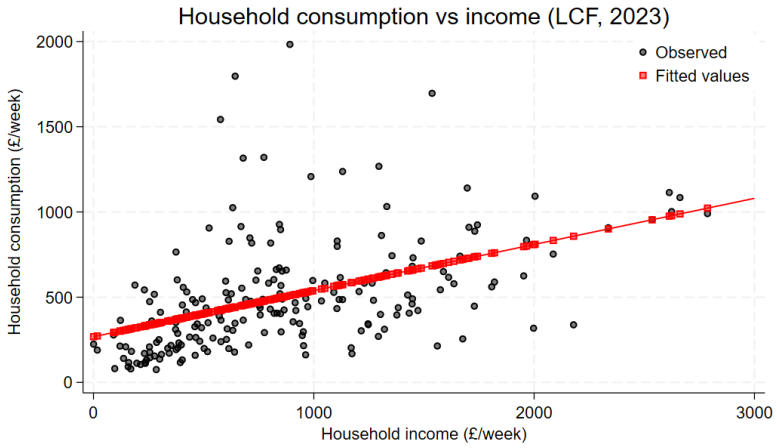
```
* Compute variance-covariance matrix
corr hhcon fitted residual, cov
```

```
(obs=200)

             |    hhcon   fitted residual
-------------+---------------------------
       hhcon |   108747
      fitted |  24333.5  24333.5
    residual |    84414 -.000076    84414
```

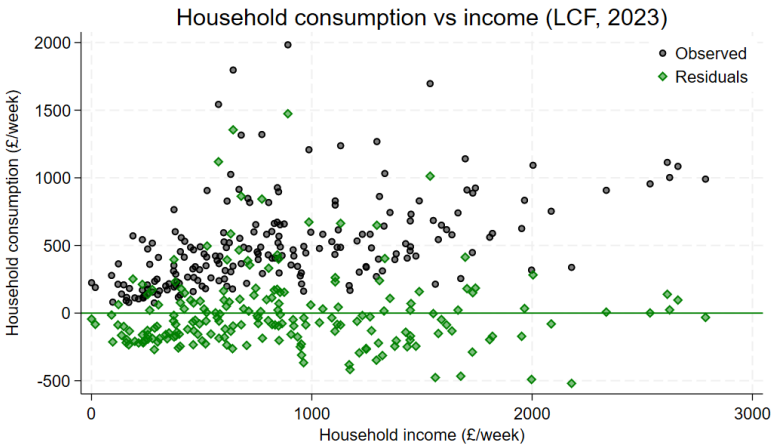
Fitted values

```
* Plot
twoway (scatter hhcon hhinc, mc(black) mfc(%50)) (scatter fitted hhinc, mc(red) mfc(%50) msi
```



Residual

```
* Plot
twoway (scatter hhcon hhinc, mc(black) mfc(%50)) (scatter residual hhinc, mc(green) mfc(%50)
```



Binary regressor:

What happens if the regressor is a 'dummy' variable?

$$[hhcons_i, children_i]$$

where $children_i$

- $= 1$: household has children,
- $= 0$: household does not have children

Summary stats

```
* Summary table
tab children, sum(hhcon)
```

```
            |      Summary of COICOP: Total
            | consumption expenditure - children
            |              and adults
   children |        Mean   Std. dev.       Freq.
------------+------------------------------------
          0 |   481.09786   342.33194         151
          1 |    575.2546   278.91476          49
------------+------------------------------------
      Total |   504.16626   329.76875         200
```

OLS - dummy variable

```
* Estimate bivariate linear regression
reg hhcon children
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =      3.05
       Model |  327978.835         1  327978.835   Prob > F        =    0.0824
    Residual |    21312759       198  107640.197   R-squared       =    0.0152
-------------+----------------------------------   Adj R-squared   =    0.0102
       Total |  21640737.9       199  108747.427   Root MSE        =    328.09

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    children |   94.15674   53.94059     1.75   0.082    -12.21506    200.5285
       _cons |   481.0979   26.69923    18.02   0.000     428.4465    533.7492
------------------------------------------------------------------------------
```

You can show,

$$\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$$
$$\hat{\beta}_0 = \bar{y}_0$$

where,

- $\bar{y}_0$ is the mean of the outcome for observations with $x = 0$

- $\bar{y}_1$ is the mean of the outcome for observations with $x = 1$

Yields a very intuitive interpretation of the OLS estimator

Sample analogue

This is the sample analogue of

$$\begin{aligned}\beta_1 =& E[y_i|x_i = 1] - E[y_i|x_i = 0] \\ \beta_0 =& E[y_i|x_i = 0]\end{aligned}$$

Which was the interpretation of the population parameter in a model with a binary ('dummy') variable.

Multivariate models

What happens if you have two regressors

$$[hhcons_i, hhinc_i, children, hhsize_i]$$

OLS - multivariate

```
* Estimate bivariate linear regression
reg hhcon hhinc children hhsize
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(3, 196)       =     27.93
       Model |  6480977.84         3  2160325.95   Prob > F        =    0.0000
    Residual |    15159760       196  77345.7144   R-squared       =    0.2995
-------------+----------------------------------   Adj R-squared   =    0.2888
       Total |  21640737.9       199  108747.427   Root MSE        =    278.11

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .1937248   .0382801     5.06   0.000     .1182311    .2692185
    children |  -238.3538   74.84546    -3.18   0.002    -385.9596   -90.74795
      hhsize |   125.7418   27.57406     4.56   0.000     71.36187    180.1217
```

```
       _cons |   121.2957   47.91901     2.53   0.012     26.79261    215.7987
------------------------------------------------------------------------------
```

Interpretation

Interpretation follows from the fact that,

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

$$\hat{\beta}_j = \frac{\partial \hat{y}_i}{\partial x_{ij}}$$

or

$$\Delta \hat{y}_i = \hat{\beta}_j \Delta x_{ij} \quad \text{holding other } x\text{'s fixed}$$

Partialling out

The following result is called the **Frisch-Waugh Theorem** (Wooldridge 2025, 76)

The OLS estimate for $\hat{\beta}_1$ from the multivariate 'projection'

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

is the same as the OLS estimate for $\hat{\gamma}_1$ simple 'projection'

$$\hat{\tilde{y}}_i = \hat{\gamma}_0 + \hat{\gamma}_1 \tilde{x}_{i1}$$

> **Projection**
>
> Why the phrase 'projection'? One interpretation of OLS is geometric. The estimator 'projects' the outcome (Y) in the (vector) space spanned by the regressors (columns of X).
>
> $$\hat{Y} = X\hat{\beta} = \underbrace{X(X'X)^{-1}X'}_{P_X} Y$$
>
> The predicted value of Y is just a 'projection' of Y: $P_X Y$. P_X is an idempotent projection matrix. The residual then is,

$$\hat{U} = Y - \hat{Y} = Y - P_X Y = (\underbrace{I - P_X}_{M_X})Y$$

M_X is another idempotent projection matrix. And crucially, $M_X P_X = 0$: the two projections are orthogonal. So, OLS projects Y into two vector spaces: one spanned ('explained by') X and another orthogonal to that space. The predicted values and residuals are mechanically orthogonal, which means that they are perfectly uncorrelated *in the data*.

where,

- $\tilde{x}_{i1}$ is the residual from the regression of x_1 against all other regressors

$$\tilde{x}_{i1} = x_{i1} - (\hat{\rho}_0 + \hat{\rho}_1 x_{i2} + \cdots + \hat{\rho}_{k-1} x_{ik})$$

and,

- $\tilde{y}_i$ is the residual from the regression of y against all regressors *excluding* x_1

$$\tilde{y}_i = y_i - (\hat{\psi}_0 + \hat{\psi}_1 x_{i2} + \cdots + \hat{\psi}_{k-1} x_{ik})$$

Demonstration

```
* Estimate bivariate linear regression
qui reg hhinc children hhsize
predict hhinc_tilde, resid
qui reg hhcon children hhsize
predict hhcon_tilde, resid

reg hhcon_tilde hhinc_tilde
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     25.87
       Model |  1980895.01         1  1980895.01   Prob > F        =    0.0000
    Residual |  15159759.9       198  76564.4441   R-squared       =    0.1156
-------------+----------------------------------   Adj R-squared   =    0.1111
```

```
       Total |  17140654.9       199  86133.9444   Root MSE        =     276.7

------------------------------------------------------------------------------
 hhcon_tilde | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 hhinc_tilde |   .1937248   .0380862     5.09   0.000     .1186181    .2688315
       _cons |  -2.04e-08   19.56584    -0.00   1.000    -38.58418    38.58418
------------------------------------------------------------------------------
```

Goodness-of-Fit

How well did we fit the data?

We have decomposed each observation into two parts:

$$y_i = \underbrace{\widehat{y}_i}_{\text{fitted values}} + \underbrace{\widehat{u}_i}_{\text{residual}}$$

We can apply this decomposition to the *total sum of squares* (SST)

$$\text{SST} \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- The SST is a measure of the total variation in y.

We can show that (see Wooldridge 2025, 33–34),

$$\text{SST} = \text{SSE} + \text{SSR}$$

where,

- *explained sum of squares* (SSE):

$$\text{SSE} \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- *residual sum of squares* (SSR):

$$\text{SSR} \equiv \sum_{i=1}^{n} \hat{u}_i^2$$

Measure of fit

One key measure of fit then is:

$$\mathbf{R}^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

This measure captures the share of the variance in the outcome variable explained by the regressors.

- If $\mathbf{R}^2 = 1$ then regressors perfectly predict the outcome.

Chasing $\mathbf{R}^2$

Many students make the mistake of thinking that higher $\mathbf{R}^2$ is always better. Here are reasons this need not be the case:

1. In social sciences, the DGP is often very complex and we don't observe many explanatory variables.
    - given the same set of $\mathbf{x}$'s, different outcomes may have very different $\mathbf{R}^2$
2. If we care about the relationship between y and a specific x_j, we do not necessarily care about explaining all of y.
3. We can always add variables and (mechanically) increase $\mathbf{R}^2$; however, this comes at a potential cost:
    a. we *MAY* inadvertantly increase the variance of our estimates (e.g., *irrelevant* variables)
    b. we *MAY* end up adding "bad controls"

Adjusted $\mathbf{R}^2$

An alternative measure fit is the *adjusted* $\mathbf{R}^2$:

$$\overline{\mathbf{R}}^2 = 1 - \frac{\text{SSR}/(n-k-1)}{\text{SST}/(n-1)} = 1 - (1-\mathbf{R}^2) \times \frac{(n-1)}{(n-k-1)}$$

where,

- n is sample size
- k is the number of regressors $\Rightarrow k+1$ is the number of parameters estimated (including intercept).

As you increase k, you decrease $\overline{\mathbf{R}}^2$

- Creates a disincentive to add irrelevant regressors that do not also reduce SSR.

Example: LCF (2023)

```
.29948045
```

```
.28875821
```

Adding "irrelevant" variables

```
reg hhcon hhinc hhsize children month tvlicense
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(5, 194)       =     16.91
       Model |  6568404.46         5  1313680.89   Prob > F        =    0.0000
    Residual |  15072333.4       194  77692.4403   R-squared       =    0.3035
-------------+----------------------------------   Adj R-squared   =    0.2856
       Total |  21640737.9       199  108747.427   Root MSE        =    278.73

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .1911845   .0385338     4.96   0.000     .1151854    .2671835
      hhsize |    124.161   27.71213     4.48   0.000     69.50531    178.8168
    children |  -233.0088   75.21666    -3.10   0.002    -381.3562   -84.66142
```

```
       month |  -1.276233   6.234806    -0.20   0.838    -13.57294    11.02047
   tvlicense |   59.82271   56.57194     1.06   0.292     -51.7523    171.3977
       _cons |   82.86154   77.50891     1.07   0.286    -70.00675    235.7298
------------------------------------------------------------------------------
```

Does OLS work?

Are these the "right" numbers?

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     57.08
       Model |  4842359.39         1  4842359.39   Prob > F        =    0.0000
    Residual |  16798378.5       198  84840.2954   R-squared       =    0.2238
-------------+----------------------------------   Adj R-squared   =    0.2198
       Total |  21640737.9       199  108747.427   Root MSE        =    291.27

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .2705895   .0358165     7.55   0.000     .1999587    .3412203
       _cons |   268.0611   37.42847     7.16   0.000     194.2515    341.8707
------------------------------------------------------------------------------
```

Let's evaluate the question:

1. it pre-supposes that there is a *right* answer (i.e. a correct estimate)
    - i.e., a *true* population parameter (β)
2. it implies a notion of "correctness"
    - do we mean $\beta = 0.27$?
    - do we mean $\beta \approx 0.27$?
    - do we mean β is 'close to' 0.27?
3. it assumes that the OLS estimator can be *right*.

First off, let's clarify something:

$$\hat{\beta}$$

- the estimator

...is a random variable

$$0.27$$

- the estimate

...is just a number

Accuracy

Is $\hat{\beta} = \beta$?

- No!
- In fact, the statement does not make sense since $\hat{\beta}$ is a random variable while β is a non-random constant.
- If we are accurate, we can ask: $Pr(\hat{\beta} = \beta) = ?$
- The answer is: $Pr(\hat{\beta} = \beta|X) = 0$ for any u with a continuous distribution (conditional on X: the sample of values of independent regressors).

Random variable

In what sense is $\hat{\beta}$ a random variable?

- The realized value of $\hat{\beta}$ depends on the realized sample[1]
- In another 'state of the world', the sample may have realized differently

[1] Recall, if u is independent of $\mathbf{x}$, then we can treat $\mathbf{x}$ as fixed. Therefore, the randomness (across samples) is really from different realizations of u. However, any discussion of u and $\mathbf{x}$ pre-supposes a model which we have not yet defined.

Example: LCF (2023) - Version 2

I randomly sampled 200 observations from the LCF (2023) data, but could have equally have sampled a different 200 observations

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     90.01
       Model |  5283929.28         1  5283929.28   Prob > F        =    0.0000
    Residual |  11623093.9       198  58702.4945   R-squared       =    0.3125
-------------+----------------------------------   Adj R-squared   =    0.3091
       Total |  16907023.2       199  84959.9155   Root MSE        =    242.29

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .2712055   .0285857     9.49   0.000     .2148341     .327577
       _cons |   232.6529   30.55968     7.61   0.000     172.3886    292.9171
------------------------------------------------------------------------------
```

Example: LCF (2023) - Version 3

... and again,

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =    102.09
       Model |  7462064.65         1  7462064.65   Prob > F        =    0.0000
    Residual |  14472533.5       198  73093.6037   R-squared       =    0.3402
-------------+----------------------------------   Adj R-squared   =    0.3369
       Total |  21934598.2       199  110224.111   Root MSE        =    270.36

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .3461268   .0342567    10.10   0.000      .278572    .4136816
       _cons |   189.5292   36.88368     5.14   0.000      116.794    262.2645
------------------------------------------------------------------------------
```

Example: LCF (2023) - Version 4

... and again.

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     27.79
       Model |  3538499.21         1  3538499.21   Prob > F        =    0.0000
    Residual |  25213603.4       198  127341.431   R-squared       =    0.1231
-------------+----------------------------------   Adj R-squared   =    0.1186
       Total |  28752102.6       199  144482.928   Root MSE        =    356.85

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |    .241712   .0458536     5.27   0.000     .1512879    .3321361
       _cons |   267.7737   48.63081     5.51   0.000     171.8729    363.6745
------------------------------------------------------------------------------
```

Can we **expect** – *a priori* – for $\hat{\beta}$ to be "on target"?

- This is a question of *biasedness*: is $E[\hat{\beta}] = \beta$?
- The answer: it depends!
- Specifically, it depends on the underlying **model**!

> **Note**
>
> It is an all to common mistake to conflate the estimator - in particular, the OLS estimator - with the linear regression model. People will say, "We estimate an OLS regression model". This doesn't really make sense, since you OLS is the estimator, not the name of the model. You can say, "We estimate a linear regression model using OLS".
> This may seem pedantic, but it is an important distinction since there are a range of estimators that can be used to estimate a linear regression model; all with different properties. In addition, the OLS estimator can be applied to any data, regardless of the true model. So it is a good idea to make sure your language is accurate.

References

Bibliography

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."